B. GCD Problem

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 900 | Tags: brute force, constructive algorithms, math, number theory |
https://codeforces.com/problemset/problem/1617/B

Given a positive integer n. Find three **distinct** positive integers a, b, c such that $a + b + c = n$ and $\gcd(a, b) = c$, where $\gcd(x, y)$ denotes the greatest common divisor (GCD) of integers x and y.

Input

The input consists of multiple test cases. The first line contains a single integer t ($1 \le t \le 10^5$) — the number of test cases. Description of the test cases follows.

The first and only line of each test case contains a single integer n ($10 \le n \le 10^9$).

Output

For each test case, output three **distinct** positive integers a, b, c satisfying the requirements. If there are multiple solutions, you can print any. We can show that an answer always exists.

Example

Input
6 18 63 73 91 438 122690412

Output
6 9 3 21 39 3 29 43 1 49 35 7 146 219 73 28622 122661788 2

Note

In the first test case, $6 + 9 + 3 = 18$ and $\gcd(6, 9) = 3$.

In the second test case, $21 + 39 + 3 = 63$ and $\gcd(21, 39) = 3$.

In the third test case, $29 + 43 + 1 = 73$ and $\gcd(29, 43) = 1$.